

Sajal Kanwal

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TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript (ES6+), Java, SQL (PostgreSQL, MySQL)
Frontend	React.js, Next.js 14 (App Router), React Query, Tailwind CSS, Zustand, Recharts
Backend & DBs	Node.js, PostgreSQL, Prisma ORM, SQLAlchemy, REST APIs, tRPC
AI Engineering	Multi-Agent Systems, LangChain, RAG Architecture, Vercel AI SDK, Semantic Search, Vector DBs
DevOps & Tools	Git/GitHub, Docker, Vercel, Stripe Webhooks, Octokit, Clerk Auth, n8n
Core Competencies	SaaS Architecture, Modular ETL Pipelines, LLM & RAG Integration, Database Query Optimization, Asynchronous Processing, Agile Development

PROJECTS

Repo Lens: AI-Powered GitHub Collaboration SaaS 

Mar' 2026

Next.js 15 · PostgreSQL · LangChain · tRPC · Google Gemini · Vercel AI SDK

A multi-tenant SaaS platform that semantically indexes GitHub repositories and exposes a streaming AI chat interface for codebase Q&A and commit intelligence.

- Designed a production-grade RAG pipeline using LangChain and Google Gemini, generating 768-dimensional codebase embeddings in PostgreSQL via `pgvector`; raw SQL cosine similarity queries (threshold >0.5) inject the top-*k* relevant files into the LLM context window, achieving sub-100ms semantic retrieval latency.
- Engineered an event-driven async processing engine via Octokit to poll Git commit diffs and trigger AI summarization jobs outside the serverless execution boundary, eliminating cold-start timeout failures for compute-heavy workloads.
- Enforced end-to-end type safety across client, server, and database layers via tRPC, Prisma ORM, and React Query; designed a repository-size-tiered credit ledger with Stripe Webhooks, enabling real-time balance reconciliation and fraud-resistant monetization.

Sentinel: Autonomous AI Content Intelligence System 

Jan' 2026

Python · PostgreSQL · OpenAI API · Docker · SQLAlchemy · n8n

A self-hosted multi-agent news intelligence pipeline that ingests, ranks, and delivers personalized daily digests — fully containerized for zero-drift deployment.

- Architected a pluggable scraping framework using the Registry Design Pattern to orchestrate heterogeneous collectors (YouTube transcripts, RSS feeds), reducing time-to-onboard-new-source to under 30 minutes and cutting manual research overhead by 98%.
- Designed a three-agent LLM pipeline (Curator → Digest → Email) with profile-weighted relevance scoring, surfacing high-signal content at 3.2× the precision of keyword-only filtering baselines.
- Shipped an idempotent email delivery pipeline via SMTP and Cron scheduling with DB-level content hashing, guaranteeing exactly-once delivery semantics across all daily digest executions.

Spendora: Real-Time Financial Telemetry Dashboard  

Jul' 2025

Next.js 14 · PostgreSQL · Prisma · Clerk · Zod · Recharts

A full-stack personal finance platform with real-time transaction tracking, analytics dashboards, and schema-enforced type safety across the entire data pipeline.

- Architected a zero-client-bundle data layer using Next.js 14 Server Components and React Suspense boundaries, improving First Contentful Paint by 40% (1.2s → 0.72s) on cold loads.
- Eliminated full-table scan bottlenecks on 10K+ records by designing composite B-Tree indexes on `(user_id, date, category)`, driving p95 query latency from >800ms to under 200ms — a 4× throughput improvement.
- Enforced schema-first type safety via Prisma and Zod across all API boundaries, eliminating runtime serialization errors and closing SQL injection vectors at the ORM layer.

CERTIFICATIONS

• Oracle Cloud Infrastructure 2025 Certified Developer Professional — Oracle	Nov' 2025
• Oracle Cloud Infrastructure 2025 Certified Generative AI Professional — Oracle	Oct' 2025
• Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate — Oracle	Oct' 2025

ACHIEVEMENTS

- Top 50 – GeeksforGeeks Hack the Future: Engineered a RAG pipeline for 10K+ docs, cutting retrieval latency by 70% (500ms → 150ms) via Recursive Character Chunking and local Qdrant (HNSW).

Apr' 2025

EDUCATION

Lovely Professional University	Phagwara, Punjab
B.Tech in Computer Science & Engineering (Data Science & ML) - CGPA: 8.55/10.0	Aug' 2023 – Present